

MALAVIYA MISSION TEACHER TRAINING CENTRE KANNUR UNIVERSITY

Thavakkara, Civil Station. P.O., Kannur District, Kerala - 670 002

ONLINE REFRESHER COURSE ON **Mathematical Foundations** of Contemporary Scientific Research

15-July-2026 to 28-July-2026

Programme Philosophy & Objectives

The central philosophy of this refresher course is to present mathematics as a foundational language of modern science and technology, viewed from the perspective of a broad scientific audience rather than that of specialists alone.

The primary objectives include:

- Exposing participants to important mathematical ideas, themes, and developments that significantly influence contemporary scientific research.
- Ensuring concepts remain accessible to faculty members from diverse academic backgrounds.
- Emphasizing intuition, motivation, practical applications, historical development, and cross-disciplinary connections with science and technology over strictly technical details and rigorous proofs.
- Helping participants appreciate the overarching beauty, power, and relevance of mathematics across various disciplines.

Target Audience

Faculty members from the following streams are eligible and encouraged to apply:

- Mathematics
- Statistics
- Physics
- Chemistry
- Computer Science
- Other allied Science and Technology disciplines

Course Structure & Key Features

- **Total Duration:** 12 working days
- **Daily Schedule:** 4 daily sessions
- **Session Length:** 90 minutes per session
- **Career Advancement:** Considered for CAS
- **Registration Fee:** None (No Registration Fee)
- **Mode of Conduct:** Online

Registration & Contact Details

Registration Portals:

<https://mmc.ugc.ac.in/>
<https://mmttc.kannuruniversity.ac.in/>

Programme Director:

MMTTC Kannur University
Phone: 0497-2700368, 9446353952
Email: ugchrde@kannuruniv.ac.in

Confirmed Speakers & Resource Persons

Note: Lecture topics and exact schedules for each speaker will be updated continuously as they are finalized. Expected format for speakers is one or two 90-minute expository sessions.

Speaker	Title / Topic	Sessions
1. K N Raghavan	Constructions using Ruler and Compass	2
2. A K Nandakumaran	<i>To be updated</i>	2
3. Krishna Hanumanthu	Connecting the Dots: The History, Geometry, and Applications of Interpolation	2
4. Kaushal Verma	<i>To be updated</i>	2
5. Ajith Parameswaran	1. Modern Astronomy: From Galileo to Gravitational Waves 2. Bayesian Inference and the Scientific Method	2
6. Jaydeb Sarkar	1. A crash course in calculus 2. A panorama of linear operators: past, present, and future	2
7. Sanjay Parui	<i>To be updated</i>	1
8. Muruganandam V	The role of groups in understanding nature	2
9. Venkatesh	Different shades of infinity	2
10. Prasuna Bandi	<i>To be updated</i>	2
11. Lakshmipriya	<i>To be updated</i>	2
12. Rajula Srivastava	<i>To be updated</i>	1
13. Annapareddy Tulasi Ram Reddy	<i>To be updated</i>	2
14. P. Jishnu Sai	<i>To be updated</i>	1
15. Abhishek Ajaykumar	How Linear Algebra Powers Machine Learning	1
16. Nikhil P A	<i>To be updated</i>	1
17. Aswin Govindan Shery	<i>To be updated</i>	2
18. Sayantan Chakraborty	The Mathematics of Vision: From Pixels to Neurons and Beyond	2
19. Joby K Jose S	<i>To be updated</i>	2
20. Pranav Haridas (KSoM)	<i>To be updated</i>	2
21. Renjith T (Govt College Kasaragod)	<i>To be updated</i>	2

Programme Schedule

[Detailed day-by-day session schedule to be prepared and inserted here.]